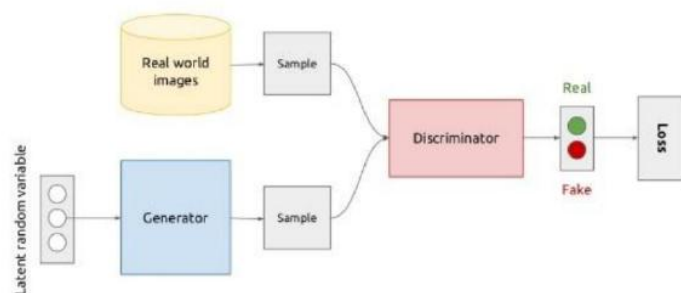


UNSUPERVISED REPRESENTATION LEARNING WITH DEEP CONVOLUTIONAL GENERATIVE ADVERSARIAL NETWORKS

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Abstract - Found out that GAN(판별기와 생성기가 경쟁하면서 학습) using CNN(이미지를 단계적으로 학습), called DCGAN is a strong candidate on unsupervised learning and also its learned features can be used on general image representations.



- GAN의 모습

Introduction – In GAN, can reuse intermediate representations(features) on supervised tasks.

It has lack of cost functions, but unstable -> propose a set of constraints on the architectural topology of Convolutional GANs (Which is called DCGAN)

Generator has a vector arithmetic property that makes manipulation of semantic qualities(의미론적 특성) easy. (재학습 없이 조작이 손쉽게 가능하다 -> 안경 쓴 남자에서 남자를 빼고 여자를 대입해서 안경 쓴 여자를 형성 할 수 있음)

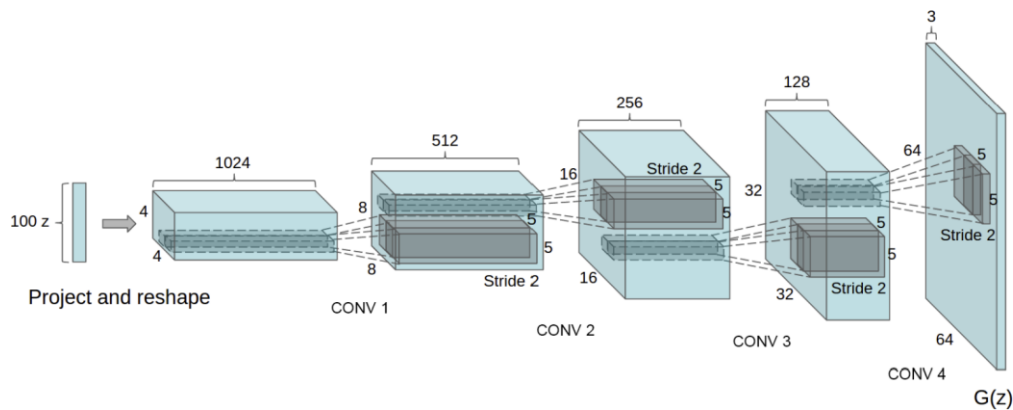
However, generating natural images of the real world have had not much success until recently.

APPROACH AND MODEL ARCHITECTURE – Had three changes to demonstrated CNN architecture

1. Used all convolutional net (maxpooling 같은 spatial pooling function을 대체) – 스스로 학습하도록 함
2. Eliminate Fully connected layer, but only connect the highest convolutional features to generator and discriminator. **여기 질문**
3. Use Batch normalization to prevent gradient vanishing -> But not applying

batchnorm to the generator output layer and the discriminator input layer. (g의 output에 사용시 결과물이 다 normalize됨, d의 input에 사용시 input들이 진짜와 가짜로 나뉘어야하는데 합쳐져버림, oscillation and model instability 발생)

4. Generator - RELU와 output layer – Tanh, Discriminator - Leaky RELU



100 -> 1024*4*4 -> 512*8*8 -> 256*16*16 -> 128*32*32 -> 3*64*64 -> G(z), Only Reshaping and Transposed Convolution are used on generating image.

To avoid memorization, Figure 1 (simple memorization) and Figure 2 (fully trained) are provided. (같은 noise가 중복적으로 발생하여 underfitting의 모습을 보임 → 이는 학습을 했다는 증거가 된다)

To remove duplicates, fit a 3072-128-3072 de-noising dropout regularized RELU autoencoder on 32x32 downsampled center-crops of training examples. The resulting code layer activations are then binarized via thresholding the ReLU activation which has been shown to be an effective information preserving technique (Srivastava et al., 2014) and provides a convenient form of semantic-hashing, allowing for linear time de-duplication. **여기 질문** -> downsampling을 통해서 전체구조만 남김?

EMPIRICAL VALIDATION OF DCGANS CAPABILITIES – When using DCGAN for supervised tasks, accuracy was higher and the number of features was lower compared to K-Means. -> **채널관련 부분 질문하기**

Since the model was not trained on the target dataset, it demonstrates domain robustness (한 도메인에서 한 학습이 다른 도메인에도 잘 작동함).

When evaluated on SVHN (적은 라벨과 데이터수로 얼마나 잘 분류하는지) (이미 다른 데이터로 학습 후 제공된 데이터로 추가학습), it achieved the lowest error rate.

Smooth transitions in the latent space (숨겨진, 잠재적인 space) during room interpolation indicate no memorization (암기가 아니다 Fig. 4).

The discriminator showed effectiveness in object detection. (Fig. 5에서 object 부분에 discriminator가 반응하는 것을 볼 수 있다)

To verify that the generator recognizes and draws major scene components → "window" filters were identified and removed (이게 창문임을 판단하고 그리는데 영향을 끼친다는 가정하에), the generated images then show that the generator "forgets" to draw windows. (이것도 학습이라는 증거) – Fig. 6

Fig. 7 - Vector arithmetic(벡터연산) can be observed.

안경 쓴 남자 – 안경 안쓴 남자 + 안경 안 쓴 여자 -> 안경 + 안경 안 쓴 여자 -> 안경 쓴 여자

Fig. 8 – A "turn vector" is demonstrated. (왼쪽 얼굴, 오른쪽 얼굴을 제공하고, 보강 작업(interpolation)을 거치면 다른 포즈의 얼굴을 생성할 수 있다)

Both results suggest further research is needed.

EVALUATING DCGANS CAPABILITY TO CAPTURE DATA DISTRIBUTIONS

KNN classification was used to evaluate DCGAN's ability to analyze and learn data distributions. DCGAN was trained on handwritten digit images and instructed to draw given digits → KNN was used to classify the generated digits → If KNN correctly classifies them, the generated images are considered high quality. Removing scale and bias parameters from batchnorm was found to improve accuracy.

Scale과 bias를 제거하니 정확도가 올라간다는걸 확인 **질문**

데이터의 변형이 아닌 생성을 한다는 사실을 알 수 있다.

질문

실제 데이터보다 정확도가 높다는 건 오히려 부정적인 상황이 아닌가?